

Maternal Health Prediction:

Can routinely collected antenatal variables be used to accurately predict maternal pregnancy risk classification?

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Code First Girls +Masters in AI/ML

Assignment 1



1. Background

Failings in maternal care have been extensively reported in the media, and most recently documented in the Amos report (Amos, 2026). My interest in this topic is driven by my personal experience of maternity care, which highlighted the challenges of fragmented services and reinforced the importance of understanding factors associated with maternal risk. There have been neonatal and maternal deaths that the Amos report identifies as avoidable, due to poor or inconsistent care.

Routine antenatal care generates large quantities of clinical data, including maternal characteristics and observations such as blood pressure, fetal heart rate and more. Where risk classifications are already recorded but the underlying decision process is not fully documented, explainable machine learning also offers a route to interrogating that process itself. It is particularly important in clinical context that it is explicitly clear which factors are driving existing risk categorisation, and with what relative weight. This has value both for prediction and for scrutiny of the classification systems currently in use.

The NHS reports substantial variation in delivery methods and maternal outcomes across age groups, deprivation quintiles and ethnic groups. Better understanding of how higher risk pregnancies are currently identified could help support earlier intervention and thus impact outcomes.

This project therefore investigates whether routinely collected antenatal information can be used to predict maternal pregnancy risk classification using supervised machine learning techniques, while comparing with trends observed in national NHS maternity statistics.

2 Research Question and Analysis Objectives

This analysis is focused on prediction in maternal healthcare, to support early intervention in pregnancies identified as higher risk, allowing for more targeted antenatal care. The main research question is therefore:

Can routinely collected antenatal variables be used to accurately predict maternal pregnancy risk classification, and which features contribute most strongly to these predictions?

In order to answer this question, the following objectives are the focus of the analysis:

1. Explore relationships between antenatal variables and reported risk classification.
2. Develop supervised machine learning models to predict risk classification.
3. Examine cases of disagreement between model predictions and reported risk levels, to identify patterns in individual-level misclassification.
4. Investigate feature importance using explainable AI techniques.
5. Compare patterns identified in the individual-level data with national NHS maternity statistics, to assess consistency for features not directly implicated in the risk classification rule.

It is important to note that in healthcare, explainability is of particular importance. Clinicians must be able to understand the basis of predicted risk in order to be able to address any risk factors appropriately. The UNESCO Recommendation on the Ethics of Artificial Intelligence (2022) identifies transparency and explainability as fundamental ethical principles, emphasising that AI systems should provide understandable and traceable explanations appropriate to their context of use. In healthcare, where decisions can directly affect maternal and neonatal outcomes, explainability is therefore essential for supporting clinician trust, accountability and appropriate adoption of machine learning systems.

3. Datasets

3.1 NHS England Aggregated Dataset

The NHS England dataset is the most recently published national maternity statistics (2024/25), and provides valuable contextual information regarding maternity outcomes in England, including stratification by deprivation and ethnicity. This stratification is directly relevant to the research question: national reporting (Amos, 2026; RCOG, 2026) identifies substantial disparities in maternal outcomes by ethnicity and deprivation, and comparing patterns from the individual-level model against these documented national trends provides a check for features not directly implicated in the individual dataset's risk classification rule. It contains records of all admissions, appointments and attendances for patients admitted to NHS hospitals in England. This includes details about types of delivery (e.g. spontaneous, Caesarean, etc.), age of mother, medical details such as birth and delivery complications, as well as the contextual information such as deprivation level and ethnicity of the mother.

3.2 Individual Maternal Health and High Risk Dataset

Publicly available, individual-level maternal health datasets with a documented risk classification are limited, given the sensitivity of patient data. Kaggle, UCI Machine Learning Repository, and Mendeley Data were searched for appropriate datasets. The dataset ultimately selected (Ahmed, 2021) was available on both Kaggle and UCI, indicating some degree of established use within the research community. This dataset included individual patient records, routinely collected antenatal variables, and a labelled risk outcome suitable for supervised classification.

The dataset (Ahmed, 2021), consists of 1,200 anonymised patient records containing routinely collected antenatal variables, including maternal age, blood pressure, BMI, diabetes status, heart rate and previous complications. These variables provide a suitable basis for supervised machine learning models aimed at predicting pregnancy risk classification with readings taken at regular intervals in routine appointments.

However, this publicly available dataset lacks detailed clinical definitions for several variables, which reduces interpretability. For mental health and pre-existing complications, the data is in the form of binary dummy variables, with no further contextual information. It is probable that the form of pre-existing complication, and mental health condition, would impact meaningfully on the actual real risk level at a more granular level.

The dataset also reports a risk level as high or low for each patient, which is reported to be based on various of these measured factors. The dataset includes a binary risk classification ("high" or "low"), which serves as the target variable for the supervised learning models developed in this project.

The exact methodology used to derive the published risk classifications is not documented, but the exploratory analysis below strongly suggests that risk level is rule derived directly from gestational diabetes and other variables. Consequently, this project evaluates the ability of machine learning models to reproduce the available classifications rather than attempting to establish new clinical risk criteria.

4. Data Preprocessing and Exploratory Analysis

The data was split on a stratified basis, based on 'Risk Level'. As supervised learning requires a known outcome for both training and evaluation, these records could not be used for modelling.

4.1 Missing Risk Level Data Analysis

There is some missing data in the individual maternal health dataset. While the proportion overall is low, and per column is low, with the highest proportion of missing values being less than 1.5%, next steps are to interrogate the type of missingness.

The BMI/Risk Level having the same number of missing values merited investigation. However, only four of these were in common. This is above what would be expected at random. While BMI is not specifically mentioned as being used to calculate risk level, it is possible that this structural association is present.

Risk Level missingness is therefore not missing completely at random (MCAR). It is significantly associated with missingness in its own likely input variables (such as blood pressure and blood sugar readings) as well as more broadly with incomplete records generally.

As risk level is a categorical variable, and the target variable, it is impossible to complete a model based imputation. The percentage of rows with missing values is low. Analysis confirmed that complete-case deletion doesn't distort feature means, std devs, or class balance, and therefore rows with missing values were dropped. After this analysis of missingness, and the dropping of these records, the training/validation/test split was conducted.

4.2 Splitting the Data

There is a roughly 60/40 split in Risk level (Low/High). It is thus important that we stratify the split by risk level to ensure consistency in each dataset. Given the uneven weighting, it is also more important to assess precision and recall within each risk level as well as accuracy. In health contexts a missed high-risk case (false negative) is a more costly error than a false positive. Class weighting will therefore be applied during modelling to ensure the high risk class is not underweighted during training.

4.3 Feature Missingness

Following the removal of records with missing risk level (Section 4.1), the remaining training set was assessed separately for feature-level missingness. As can be seen in Table 1, the training set contained low levels of missingness across the predictors, with the highest proportion occurring for BMI (10 of 711 records), while all other variables contained fewer than 0.5% missing values.

Table 1: Counts of Missing Values for each feature in the individual patients maternal health training dataset

Feature	Count of Missing Values
Systolic BP	3
Diastolic	1
BS	1
BMI	10
Previous Complications	1
Preexisting Diabetes	1

To determine whether complete-case analysis would introduce bias, feature distributions before and after removing incomplete records were compared. Only 15 records were removed, which constitutes 2.1% of the dataset. Both the feature distributions and class balance of risk level were effectively preserved, and thus complete-case analysis was considered appropriate for this dataset. The same complete-case rule was subsequently applied to the validation and test datasets.

4.4 Outliers

Figure 1 shows the boxplots of numeric features. This shows a number of outliers to be investigated, and also that one BMI reading in the training dataset is zero. On further investigation, the patient with a BMI reading of 0 had readings that were otherwise plausible. This value is therefore treated as missing and imputed from the median value for BMI in the training dataset.

Figure 1 and Figure 2 show right skew distributions in Age, BS and Body Temp. IQR-based thresholds classified 14–15% of observations as outliers for BS and Body Temp, suggesting that the method was not appropriate for these variables. As these extreme values are clinically plausible and may be informative for risk prediction, no observations were removed on the basis of outlier status. Log transformation improved the distribution of Age slightly but provided limited benefit for Blood Sugar and Body Temperature. Therefore, log transformation will not be used in the modelling stage.

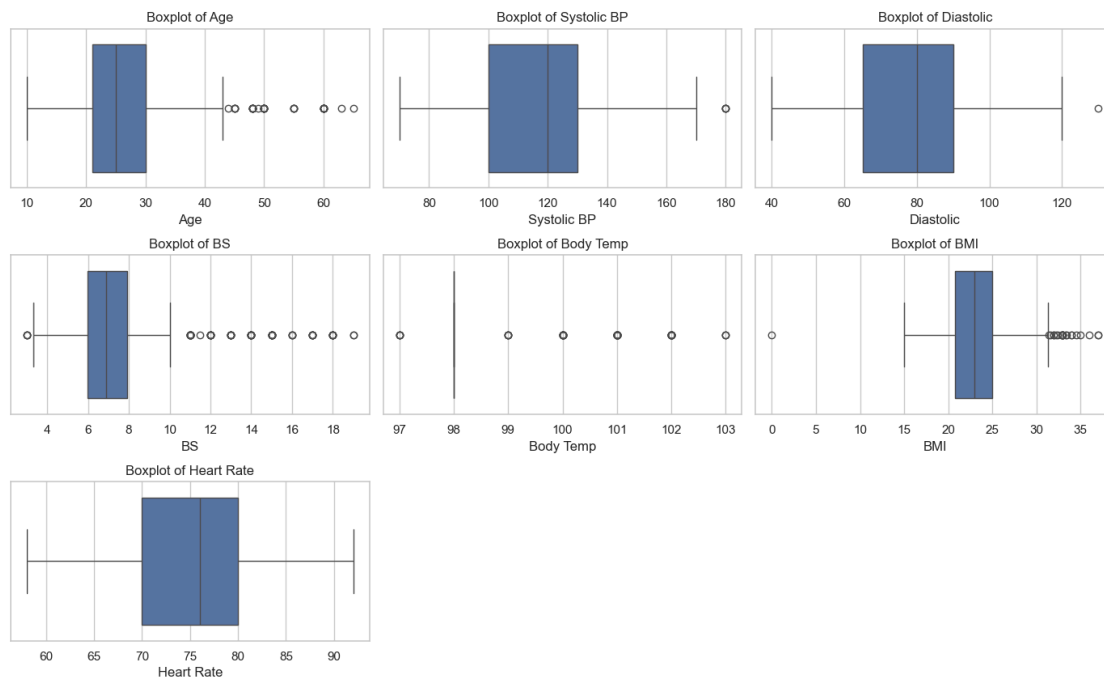


Figure 1: Boxplots showing distribution and outliers for numeric (not dummy) features

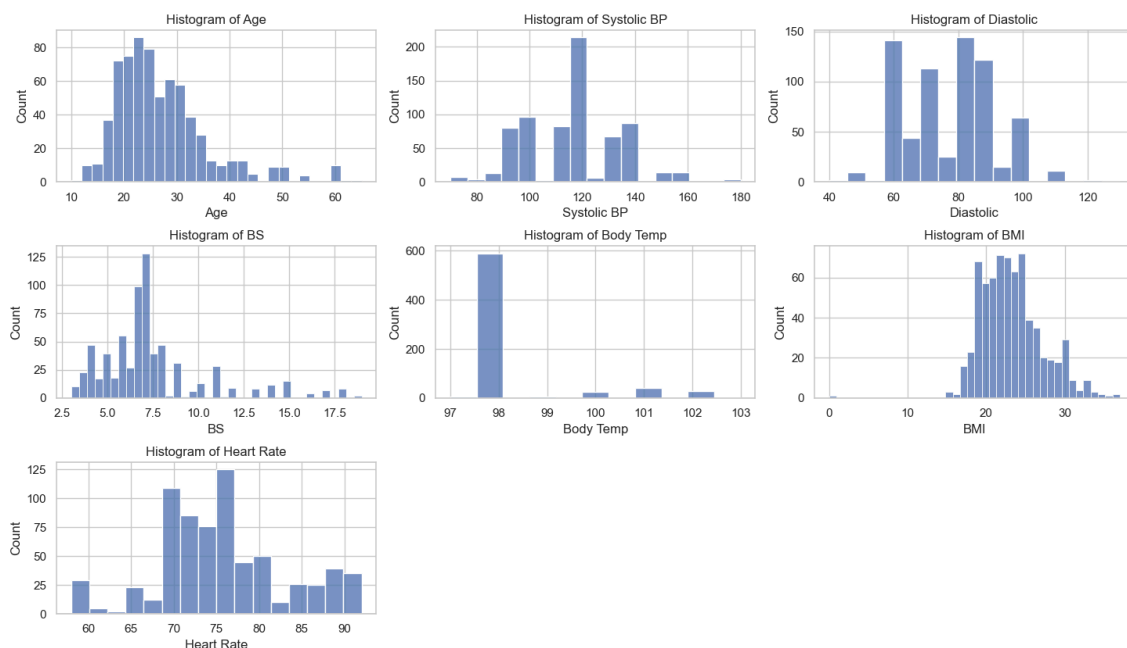


Figure 2: Histograms showing distribution for numeric (not dummy) features

4.5 Features Relationship with Risk Level

When investigating the numeric features and their relationship with risk level, Figure 3 indicates that the biggest distinction between high and low risk appears to be in blood sugar. This is clinically plausible, as diabetes (gestational or preexisting) is a known risk factor in pregnancy (NHS, 2022). Both the median and the shape of the distribution of blood sugar levels are different between low and high blood sugar.

Systolic blood pressure, diastolic blood pressure, and heart rate also show higher levels in the high risk group. Age and BMI show slightly higher values overall in the high risk group but also considerable overlap, and age has a similar distribution in both risk categories. Given the overlap across all numeric predictors, a multivariable machine learning model is likely to be more appropriate than relying on any single clinical measurement.

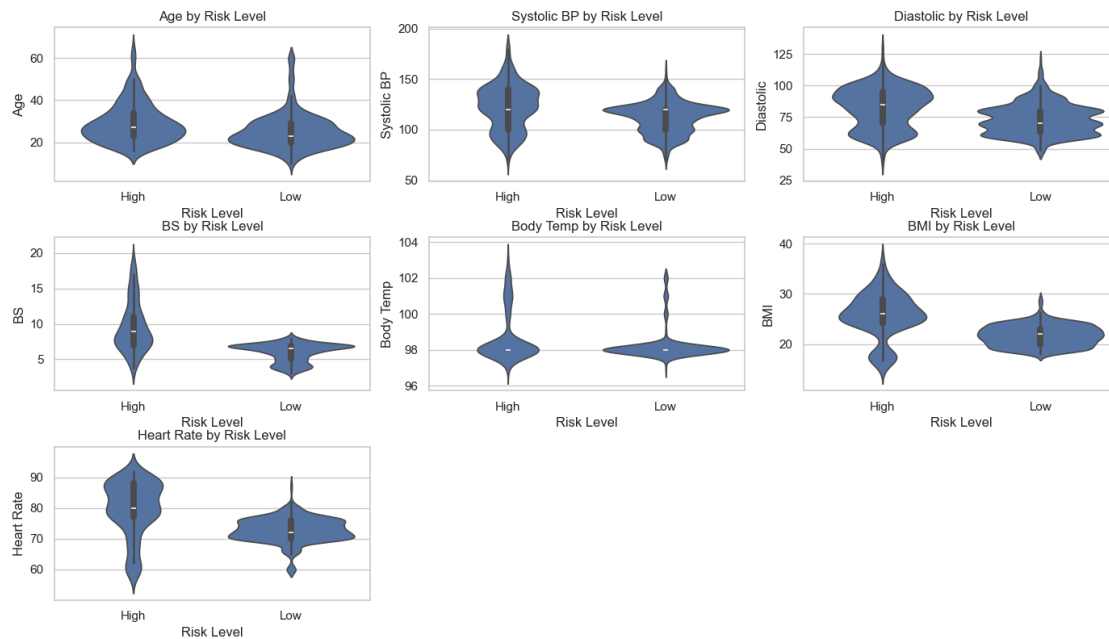


Figure 3: Violin plots showing shape of distribution of numeric features categorised by risk level

Table 2: Associations of continuous features with maternal risk level

Variable	Mann-Whitney U	p-value
Age	42199.0	1.0305e-09
Systolic BP	39724.5	6.7829e-13
Diastolic	37820.5	3.1179e-15
BS	16205.5	1.7337e-58
Body Temp	50098.5	1.1179e-06
BMI	21018.5	3.2379e-46
Heart Rate	23582.5	1.4864e-40

The binary variables, encoded as 0/1, show apparently strong associations between all these features and risk level, as illustrated in Figure 4 and Table 2. All patients with gestational diabetes are assessed as high risk, so this is a very strong association and is likely to be a direct contributor to the assessment of risk level. Not all patients without gestational diabetes are low risk, but approximately two thirds are deemed low risk. Preexisting diabetes shows a similar, although not quite as strong, association, as almost all patients with preexisting diabetes are within the high risk group. The association between previous complications and risk level is also very strong. Although mental health shows the least correlation, this still

shows an apparently strong correlation with approximately 80% of patients with a mental health condition categorised as high risk. Therefore, all these binary variables were likely used to form the original risk level assessments.

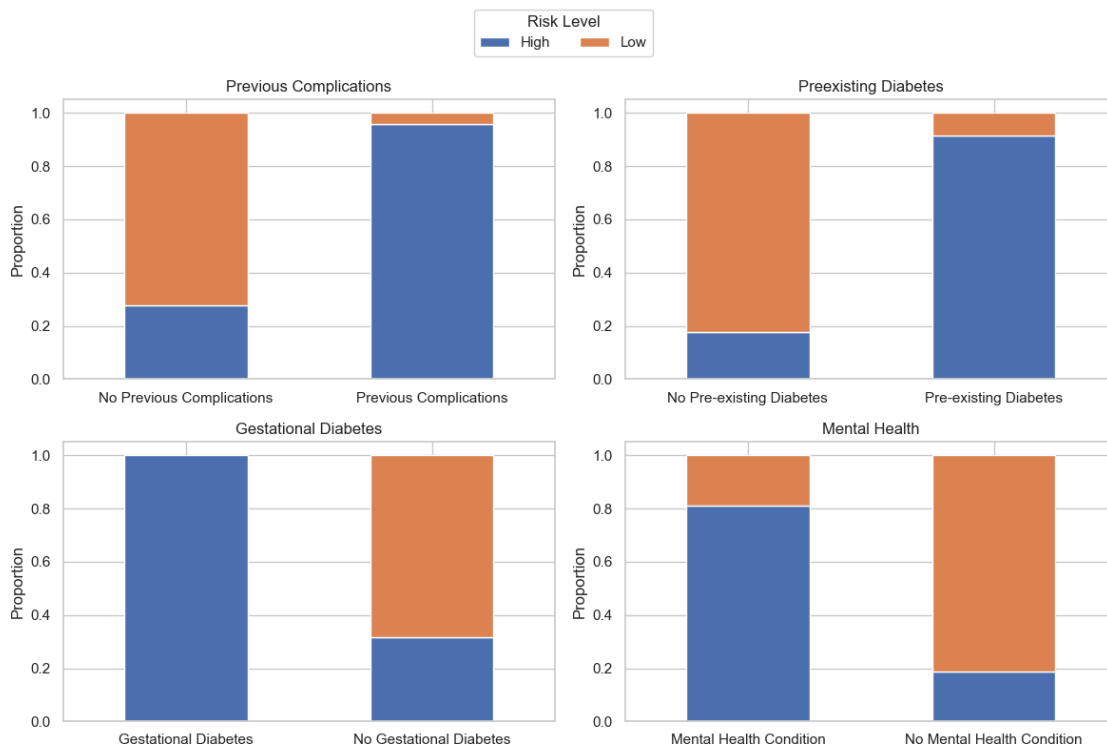


Figure 4: Stacked bar charts showing the proportion of cases deemed high risk for each of the binary features

Although these observed counts are low, the expected counts are in no cases less than 5, so a Pearson chi squared test is appropriate to further assess the relationship between these binary features and risk level. As can be seen in Table 2, all four binary features are statistically significant and are retained for model development.

Table 3: Associations of binary features with maternal risk level

Variable	χ^2	p-value
Previous Complications	195.84	1.692e-44
Preexisting Diabetes	328.72	1.828e-73
Gestational Diabetes	139.71	3.081e-32
Mental Health	252.15	8.812e-57

4.6 Features relationships with each other

When building a multivariate model, it is important to consider multicollinearity. There are a number of variables that may plausibly be strongly correlated (such as systolic and diastolic blood pressure, or BMI and blood sugar). It is possible that a model might be useful based on a more limited range of features.

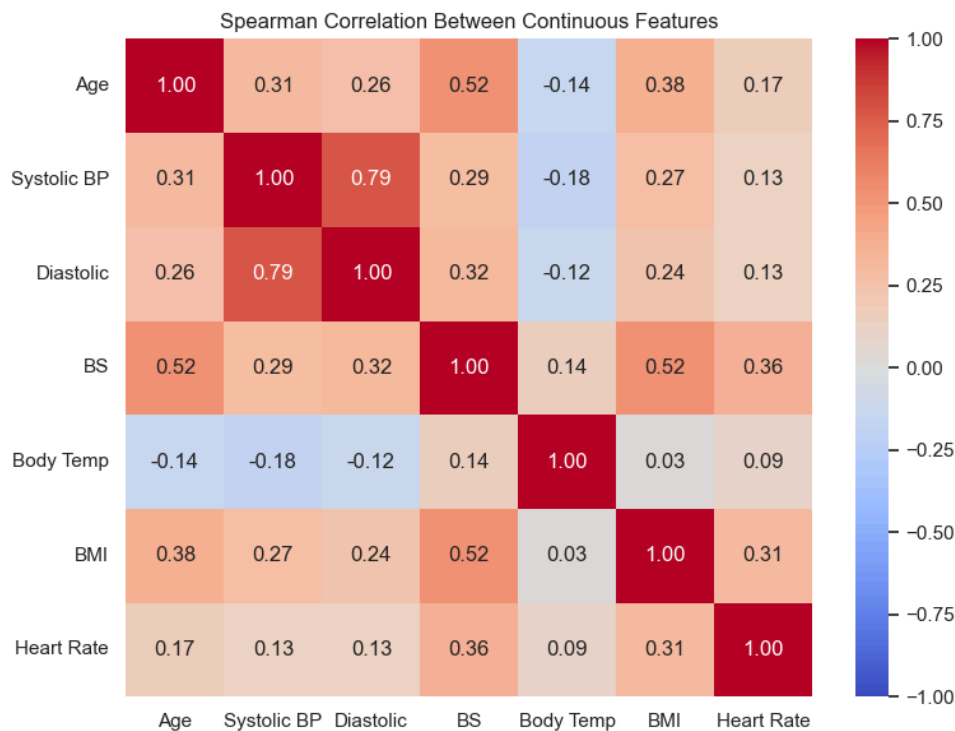


Figure 5: Correlation matrix for numeric features

As can be seen in Figure 5, there are several relationships between numeric features. Systolic and diastolic blood pressure shows strong correlation (0.79). BMI shows a moderate/strong correlation with blood sugar (0.52), while blood sugar also shows a moderate/strong correlation with age, moderate correlation with systolic and diastolic blood pressure, and with heart rate and age.

Cramér's V values for the associations between binary features were generally modest (0.03–0.43), with no pairs indicating strong multicollinearity (and thus redundancy). The strongest associations were between Mental Health and both Previous Complications ($V=0.43$) and Preexisting Diabetes ($V=0.40$), suggesting some shared information between these variables that will be examined further at the modelling stage.

For investigating multicollinearity between binary and numeric features, point biserial correlation was used (Volk-Jesussek, 2026). None of these associations were sufficiently strong to cause concern about multicollinearity, with blood sugar vs preexisting diabetes showing the strongest association ($r = 0.524$).

Thus, no features were sufficiently strongly associated with others to cause significant concern about multicollinearity and/or redundancy.

4.7 NHS Dataset Exploratory Analysis

The NHS dataset highlights some interesting points for future analysis. For example, in 2024/25 there were 92 deliveries to mothers aged 14 and under, while there were 137 births to mothers aged 50 and over. This is no doubt a changing profile over time, where average maternal age has increased. It would be of interest, for example, to investigate predictions stratified by age, as there are potentially additional confounding factors that may influence the predictive power of any ML model.

In the NHS dataset, there is also data stratified by deprivation level and ethnicity. Amos (2026) reported, for example, that Black mothers were three times more likely to die in childbirth than white women, while the rate of still births is doubled for Black babies. Therefore, adjusting the model to account for ethnicity would be an important additional step although not possible with the individual dataset which lacks ethnicity data. Equally, deprivation has a negative impact on risk level. The Royal College of Obstetricians and Gynaecologists (2026) observe that women in the most deprived areas are twice as likely to die in childbirth as women in the least deprived areas.

Therefore, an ethical and explainable ML model assessing risk level classification would ideally be stratified by ethnicity and deprivation in order to promote early identification of high risk pregnancies and focus interventions where most needed.

5. Conclusion

Exploratory analysis indicates that the Ahmed Maternal Health Dataset (2021) is suitable for constructing a machine learning model. Complete-case analysis resulted in the removal of only a few rows that did not materially affect the balance of the features or of the risk level. Statistical analyses demonstrated significant associations between all continuous and binary predictors and maternal risk level. Correlation analyses identified several moderate associations between predictors but no evidence of multicollinearity sufficiently severe to justify excluding any features from the initial model.

One of the most important findings was the near-deterministic relationship between gestational diabetes and high-risk classification, alongside strong associations for preexisting diabetes and previous complications. This suggests that risk level was likely derived directly from these variables. This project therefore evaluates the ability of ML models to reproduce the dataset's existing risk classification rather than to establish new clinical risk criteria, and feature importance results in the following stage should be interpreted with this in mind.

These findings motivate a multivariate machine learning model, as substantial overlap exists between the distributions of individual predictors and no single variable is sufficient to wholly distinguish high and low risk pregnancies. Although the individual dataset lacks contextual information such as ethnicity and deprivation levels, an appropriate basis for developing an explainable classification model whose outputs can be compared against national trends. The next stage of the project will compare supervised machine learning algorithms using the prepared training, validation and test datasets, applying class weighting to address the class imbalance identified during preprocessing, to identify a model capable of reproducing maternal risk classification while remaining interpretable and clinically relevant.

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